



جامعة الأميرة
نورة بنت عبدالرحمن



Green Goal

AI-Powered Sustainable Fan Engagement in Saudi Stadiums

graduation project

2025 – 2026



Introduction



Saudi Arabia's growing number of sports events has led to a significant increase in single-use cup waste inside stadiums, affecting both cleanliness and environmental sustainability. At the same time, fan participation in eco-friendly practices remains low due to the lack of simple, engaging solutions.

Green Goal was developed to address this challenge by encouraging the use of reusable cups through AI-based verification, GPS tracking, and a reward system that motivates fans to participate. The project aims to reduce waste, improve the match-day experience, and support Saudi Vision 2030's environmental goals.



System feature and objective



Objectives :

-  Reduce plastic waste
-  Promote sustainable behavior
-  Increase fan engagement
-  Ensure accurate verification (AI & GPS)
-  Provide data insights
-  Support Vision 2030
-  Encourage competition





System feature and objective



Features :

-  AI-based reusable cup verification
-  GPS-based location validation
-  Gamification system (points & rewards)
-  Leaderboard (individual & team competition)
-  Sustainability impact tracking
-  Reward redemption system





AI Image Classification Model



	Model: MobileNetV2 (Pre-trained on ImageNet)	
	Technique: Image Classification (Binary)	
	Classes: Reusable Cup / Not Reusable	
	Dataset: Collected from Roboflow	
	Training Images: 5,200+	
	Accuracy: 99 %	
	AI Vision: The model analyzes uploaded images to detect reusable cups using computer vision.	